

task-1

WAN Connection Comparison

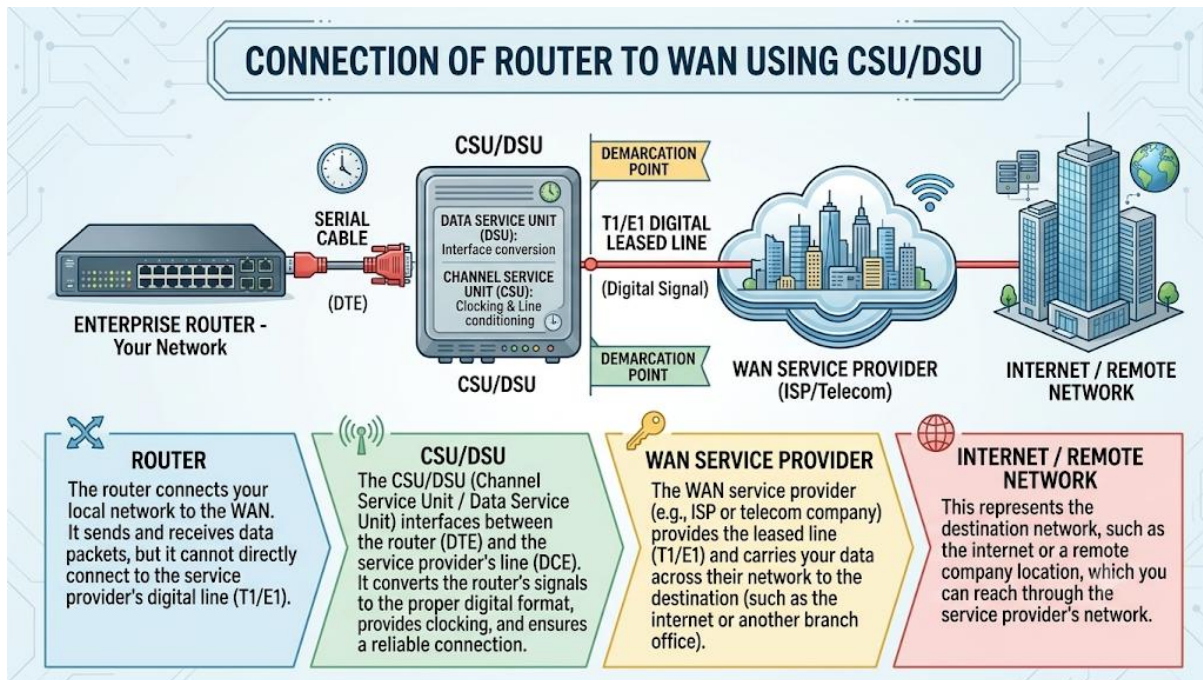
WAN Connection	Typical Speed	Cost	Typical Use Cases
Leased Line	High and dedicated; commonly 10 Mbps to multi-Gbps	High	Businesses needing dedicated, reliable connectivity between offices/data centers
DSL	Usually 1–100+ Mbps, depending on technology and provider	Low	Home users, small offices, basic internet access
MPLS	Commonly 10 Mbps to 10 Gbps+ depending on provider/service	High	Enterprises connecting multiple branches with managed, predictable private WAN connectivity
Fiber WAN	100 Mbps to multi-Gbps	Medium–High	Businesses, campuses, cloud/data-center connectivity, high-bandwidth applications

Summary:

- **DSL:** Lower cost and suitable for basic connectivity.
- **Leased line:** Dedicated connection with high reliability, but generally expensive.
- **MPLS:** Managed enterprise WAN for connecting multiple locations.
- **Fiber:** Very high bandwidth and suitable where fast connectivity is required.

Actual speeds and prices vary by ISP, location, service plan, and technology.

Task-2



Task-3

Real-World Applications Using WAN

App/Service How WAN Enables Its Functionality

IRCTC WAN connects users across India to IRCTC's centralized servers. It allows users to search trains, check seat availability, book tickets, and receive booking information over the Internet.

WhatsApp WAN/Internet connectivity allows phones to communicate with WhatsApp's servers and exchange messages, images, videos, voice messages, and calls with users in different locations.

Zomato WAN connects customers, restaurants, delivery partners, and Zomato's cloud servers. It enables restaurant searches, online orders, payment processing, order tracking, and delivery updates.

Task-4



Device: Cisco ASR 1001-X Aggregation Services Router

Key Features

1. Pay-as-You-Grow Scalable Throughput:

The router features an integrated Cisco Flow Processor / Embedded Services Processor (ESP) that delivers base forwarding throughput of 2.5 Gbps, which can be upgraded via software licenses up to 20 Gbps without needing physical hardware replacement.

2. Hardware-Accelerated Security & Traffic Management:

It includes a dedicated built-in security engine supporting up to 8 Gbps of hardware-based IPsec crypto throughput, enabling high-speed VPN encryption, Zone-Based Firewall (ZBFW), and Quality of Service (QoS) across enterprise WAN edge and branch connections.